

WASP-10b

R-2365

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_110615 · created 2026-08-19T23:51:49+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455727.93623 +/- 0.00099 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.012
Transit depth (Rp/Rs)^2	3.8 +/- 0.48 [%]
Orbital Inclination (inc)	89.71 +/- 0.64
Airmass coefficient 1 (a1)	1.28 +/- 0.018
Airmass coefficient 2 (a2)	-0.142 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	None
Optimal Aperture	3.54
Optimal Annulus	11.22
Transit Duration (day)	0.0962 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.195 $\pm$ 0.012 vs 0.1592 (deviation 2.98 σ)
Duration fit vs published	0.0962 d vs 0.0946 d (deviation 1.35 σ)
Mid-time O-C	-0.7 min
Depth SNR	16.2
Residual scatter	1.38 %

Light curve

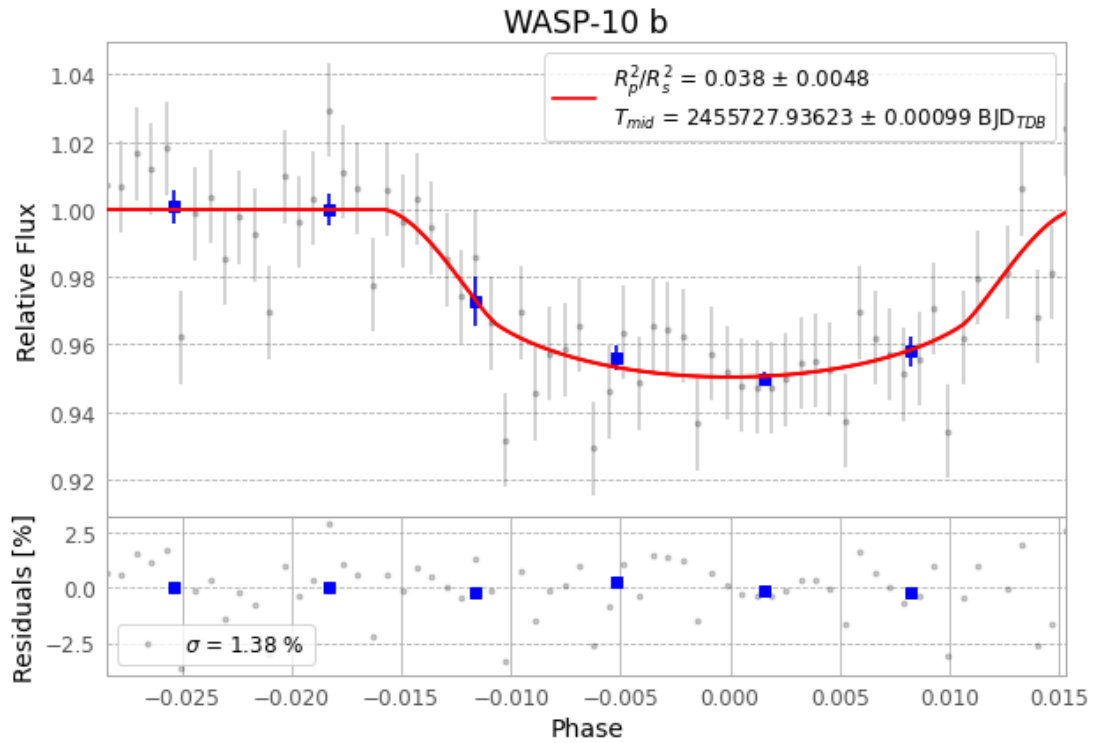


Figure 1 — FinalLightCurve_WASP-10 b_2011-06-15.png (SHA-256 cbe1619d25707474...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [221, 208], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 23b7ab1efeaf3a3f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

74 FITS frames (49 MB), WASP-10110615082113.FITS → WASP-10110615120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-110615.log (7938614212e6...), FinalLightCurve_WASP-10 b_2011-06-15.png (cbe1619d2570...), FinalLightCurve_WASP-10 b_2011-06-15.csv (bdc2e3936e2d...)

Compute resources

Wall time 456.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 23:51Z.